Course 1: Modern React 18 & Redux Toolkit Masterclass

React 18 & Concurrent Rendering Architecture Guide

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1. Understanding Concurrent Mode & Fiber Architecture

React 18 introduces concurrent rendering, allowing React to prepare multiple versions of your UI concurrently behind the scenes. This prevents expensive UI re-renders from freezing user interactions.

The Fiber tree is a mutable data structure where each node represents a component and its state. Interruptible rendering splits work into discrete chunks that yield control back to the browser event loop.

```
const [isPending, startTransition] = useTransition();

function handleSearch(term) {
  setQuery(term); // Urgent state update
  startTransition(() => {
    setFilteredResults(computeExpensiveFilter(term)); // Transition
  });
}
```

2. State Management with Redux Toolkit (RTK)

Redux Toolkit streamlines immutable state updates using Immer internally. Slices encapsulate initial state, reducers, and auto-generated action creators.

Selectors should be memoized using createSelector to avoid unnecessary recomputations when non-relevant branches of state change.

```
export const counterSlice = createSlice({
  name: 'counter',
  initialState: { value: 0 },
  reducers: {
    increment: (state) => { state.value += 1; },
    addAmount: (state, action) => { state.value += action.payload; }
  }
});
```

3. Performance Optimization Rules

Always wrap expensive callbacks passed to memoized children in useCallback.

Use React.memo only when profiling demonstrates measurable re-render bottlenecks.

Avoid instantiating inline arrow functions or object literals directly inside render JSX if the component is wrapped with React.memo.

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